Department of Economics

MSc Extended Essay Cover Sheet

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TITLE	**Financial repression: optimal and non-optimal policymaking in response to fiscal shocks**

WORD COUNT	6	0	6	8

Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered.*

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Financial repression: optimal and non-optimal policymaking in response to fiscal shocks

Candidate number: 25929

May 30, 2022

Abstract

This essay studies the optimal Ramsey policy and financial repression in a simple stochastic log-linearised New Keynesian model. The theoretical framework includes a government financing exogenous taxes and transfers with long-term nominal bonds. Two theoretical results concerning the optimal policy are derived, and the behaviour of model variables in response to a fiscal shock is analysed. Three significant findings are: First, price stability is generally more desirable than output gap stability in optimal financial repression regimes. Second, both very long and short bond durations make financial repression less "painful" from a household welfare point of view. Third, an effective lower bound can reduce welfare loss due to a fiscal shock under non-optimal policy.

1 Introduction

Following the Great Recession, the European debt crisis, and, more recently, the COVID-19 crisis, financial repression [1] and the question of reducing high government debt burdens appeared again at the forefront of economic policy debates. In addition, high inflation rates

[1] In this essay, I define *financial repression* as the reduction of the real government debt burden by the curtailment of the return on bonds, possibly through inflation, brought about by government policy.

now emphasise the question of what is the least costly way to do this - i.e. what characterizes optimal financial repression, and should it be undertaken at all?

Schmitt-Grohe and Uribe (2002) approach this problem through a New Keynesian model with non-state-contingent nominal bonds, distortionary income taxes and full commitment. In their model, fiscal shocks are accommodated through income taxes and inflation - but both price stability and labour market efficiency are essential for household welfare: the first because inflation and deflation change the relative price of firms under price stickiness compared to the efficient level of inflation, leading to a sub-optimal distribution of goods in the consumption basket; the second due to the inefficiently low level of labour hours due to the tax wedge in the labour market choice.

Then, the central authority must trade off these two instruments. Using parameter estimates based on the US economy, Schmitt-Grohe and Uribe (2002) show that, under any amount of price stickiness, their model decides the issue overwhelmingly in favour of price stability (and in favour of using income taxes to accommodate fiscal shocks). That is to say, in their model specification, financial repression is, if possible, not to be undertaken at all by a government maximizing household welfare.

Siu (2004), using a similar model allowing for both sticky and non-sticky price sectors, also calibrating the parameters of a New Keynesian model based on US data, analyzes the optimal Ramsey policy and the policymaker's reaction to a fiscal shock. He finds that the Ramsey solution prescribes essentially constant deflation, even with a low share of sticky price firms - the model predicts that increased tax collection and persistent debt accumulation is a better alternative to financial repression. However, Siu (2004) qualifies this with the observation that if government spending becomes highly volatile, inflation volatility also remains optimal to a degree in the model.

In contrast, Lustig et al. (2008) focus on optimal maturity management in hedging - and responding to - fiscal shocks. In their framework, the principal policy trade-off concerns the maturity of government bonds, and their model predicts that governments should almost exclusively rely on long-term nominal debt. The reason is that the return on long-term nominal bonds is more volatile than that on short-term bonds, creating ample space for

financial repression to respond to fiscal shocks.

This essay will look at a simple New Keynesian model. It will focus on optimal financial repression as a response to fiscal shocks, ignoring income taxes and financing alternatives to debt. The implications of this optimal policy on the model variables are one of the central focuses here.

The government sells long-term nominal bonds, and the real return on these is the target of all financial repression strategies. The central trade-off in our model is the one between output gap and inflation: this is not the same type as the one between taxes and price stability, as a movement in one implies a movement automatically in the other, but the relative change is subject to policy choice. From this point of view, I will investigate the circumstances under which it is better - from a welfare loss perspective - and to what degree to rely on inflation rather than output in accommodating shocks. Hence, my analysis will allow a Schmitt-Grohe and Uribe (2002) and Siu (2004) type evaluation of the desirability of price stability in the context of debt reduction.

In addition, we will also look at the form of the optimal Ramsey policy function, giving the response of the short-term nominal interest rate to real variables and shocks. The duration of the government debt - which can be interpreted as a proxy for maturity - is also a central part of the model, and analysing it allows an approach resembling the maturity management evaluation of Lustig et al. (2008).

The essay is structured in the following way. In section 2, I present the central model, look at a theoretical presentation of the inflation - output gap trade-off, and discuss my model's optimal Ramsey policy function. Next, in section 3, I present the parameter calibration used in the simulations in section 4. There, I analyse the response of variables under the baseline case and different parameter values, assuming optimal policy. I also look at the special case of persistent fiscal shocks and the implications of the existence of an effective lower bound, assuming the government does not follow the optimal policy. Finally, I conclude.

2 Theory

This section presents the model used in this paper and two important results regarding the policymaker's behaviour. First, I will discuss the log-linear model equations used in section 4. Next, I will present a result concerning the inflation - output gap trade-off that the policymaker faces. Finally, I will present the optimal Ramsey policy function of our model.

2.1 The baseline model

Our baseline model is given by Sheedy (2022): it is a simple log-linearized New Keynesian model augmented with a government sector[2] and fiscal policy. Besides the standard equations for the representative household and the firms - which I will not discuss -, we assume that the government finances itself by selling long-term bonds with nominal price Q equal to:

$$Q_t = \frac{1}{1+j_t} + \frac{\delta}{(1+j_t)^2} + \frac{\delta^2}{(1+j_t)^3} + ... = \frac{1}{1+j_t-\delta}$$

In each period t, where j_t is the yield-to-maturity of the bond, while $\delta \in (0,1]$ is related to the per period coupon. It can be shown that the ex-post real return R_{t+1} - in period $t+1$ - on these bonds is given by:

$$1 + R_{t+1} = \left(\frac{1+j_t}{1+\pi_{t+1}}\right)\left(\frac{1-\frac{\delta}{1+j_t}}{1-\frac{\delta}{1+j_{t+1}}}\right)$$

Where π_{t+1} is inflation in period $t+1$ relative to t. The government is assumed to have no expenditures apart from debt repayments, and collects taxes T_t in each period. Hence, the government budget constraint is:

$$T_t + D_t = (1+R_t)D_{t-1}$$

Where D_t is the real value of government debt at the end of period t. Sheedy (2022) also

[2] I use the terms government, monetary authority and fiscal authority synonymously.

defines:

$$\phi_t = \frac{\bar{T} - T_t}{\bar{D}}$$

As the fiscal shock, a mean zero white noise process, where $\bar{T}$ and $\bar{D}$ are the steady-state value of taxes and real debt, respectively. Taxes appear in our log-linearized model solely in this term, and by treating ϕ as an exogenous shock, we are implicitly treating taxes as exogenous random variables distributed around the steady-state. Hence, a positive fiscal shock represents a shortfall of taxes (or an increase in expenditure) that cannot be financed with future taxes, only through debt - thus, the need for financial repression to reduce real debt to the steady-state arises.

The expected real return on long-term bonds has to satisfy an arbitrage condition with the real interest rate on single-period bonds, r_t, defined through the Fisher equation by the short-term nominal interest rate i_t and inflation.[3] Log-linearizing and combining these gives us our baseline model equations:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa y_t$$

$$y_t = \mathbb{E}_t y_{t+1} - \sigma r_t$$

$$r_t = \mathbb{E}_t R_{t+1}$$

$$j_t = \beta\delta \mathbb{E}_t j_{t+1} + (1 - \beta\delta) i_t$$

$$R_t = j_{t-1} - \pi_t - \frac{\beta\delta}{(1 - \beta\delta)}(j_t - j_{t-1})$$

$$\beta d_t = d_{t-1} + R_t + \beta\phi_t$$

Where variables are in terms of their log-deviation from steady-state. y_t represents the output gap, while d_t is the real debt in log-deviation terms. The interpretation of the parameters in this model can be found in Table 1. Importantly, the coefficient on output gap

[3] Although the government, importantly, does not participate in short-term debt markets, I assume that there is a zero net supply of short-term bonds held by the representative household. Then, the government can set the nominal rate on these markets, by decree or law.

in the Phillips curve is:

$$\kappa = \frac{1}{1+\epsilon\eta}\left(\frac{1}{\sigma} + \frac{1+\eta}{\psi} + \eta\right)\frac{(1-\theta)(1-\beta\theta)}{\theta}$$

Where the first term is inversely related to firm-level real rigidities, the second is inversely related to aggregate-level real rigidities, and the third is inversely related to nominal rigidities.

2.2 The inflation - output gap trade-off

In contrast to Schmitt-Grohe and Uribe (2002) and Siu (2004), the policymaker in our model does not balance distortionary income taxes and inflation; rather, in each period t_0, it aims to reduce the following approximate loss function of the representative household, as argued by Sheedy (2022):

$$L_{t_0} = \frac{1}{2}\sum_{t=t_0}^{\infty}\beta^{t-t_0}\mathbb{E}_{t_0}\left[\frac{\epsilon}{\nu\xi}\pi_t^2 + \gamma y_t^2\right]$$

The policymaker aims to keep both inflation and output gap close to their steady-state (i.e. keeping the log-linear variables close to zero) through changes in its policy instrument, the short-term nominal interest rate. However, due to the functional form, a unit of change in one variable does not generally lead to the same change in welfare as a unit of change in the other. Hence, the policymaker will aim to reduce welfare loss by manipulating the change in inflation relative to the change in the output gap, $\frac{d\pi_t}{dy_t}$.

Under any given parameter calibration, and assuming that the policymaker can commit to future interest rate changes, the relative change in the variables can be modified using forward guidance, which is the key to the inflation - output gap trade-off. The following proposition gives a forward guidance trade-off function under perfect foresight in our model, ignoring the long-term bond market and the government sector.

Proposition 2.1. *Assuming perfect foresight and full commitment, and ignoring the government sector, the relative change in inflation relative to output gap, given a unit change in the*

nominal short-term interest rate m periods in the future is given by the function:

$$\left.\frac{d\pi_t}{dy_t}\right|_{di_{t+m}} = \kappa \sum_{l=0}^{m} \beta^l \frac{g(m-l)}{g(m)}; \forall t$$

Where:

$$g(m) = -\left(\frac{c_1}{r_1^{m+1}} + \frac{c_2}{r_2^{m+1}}\right)$$

$$r_1 = \frac{\frac{(1+\beta+\sigma\kappa)}{\beta} + \sqrt{\frac{(1+\beta+\sigma\kappa)^2}{\beta^2} - \frac{4}{\beta}}}{2}$$

$$r_2 = \frac{\frac{(1+\beta+\sigma\kappa)}{\beta} - \sqrt{\frac{(1+\beta+\sigma\kappa)^2}{\beta^2} - \frac{4}{\beta}}}{2}$$

$$c_1 = \frac{\frac{\sigma}{\beta} - \sigma r_1}{r_2 - r_1}$$

$$c_2 = \frac{\sigma r_2 - \frac{\sigma}{\beta}}{r_2 - r_1}$$

The proof of this proposition is found in Appendix B. The point of this exercise is to underscore that using the relevant forward guidance trade-off function - taking into account the government sector as well - the policymaker can optimally trade off inflation and the output gap by choosing the path of the nominal interest rate as a response to the fiscal shock. This feature will figure prominently in the following sections.

2.3 The optimal Ramsey policy function

The previous subsection highlights the general principle based on which the policymaker can make the optimal loss-reducing choice. This choice is, mathematically, given by a function, observed and taken into account by the representative household and the firms - I assume full commitment throughout this essay. In addition, the policymaker must take into account the market for long-term bonds, possible fiscal shocks, and, in particular, the limiting condition on real government debt.

The following proposition presents the optimal Ramsey policy function in our model.

Proposition 2.2. *Taking into account the model equations and the approximate loss function of the representative household, the optimal Ramsey policy response function is given by:*

$$i_t = \frac{1}{\beta}\Big[\frac{\kappa\epsilon(\beta\delta - 1)}{\nu\xi\gamma\sigma} + 1\Big]\pi_t - \frac{1}{\beta}\Big(\kappa - \frac{\epsilon\kappa^2 + \nu\xi\beta\delta\gamma}{\nu\xi\gamma\sigma}\Big)y_t - y_{t-1} + \Xi(L)\phi_t; \forall t$$

Where $\Xi(L)$ is a moving average lag polynomial given by the time series process of the minimization problem's Lagrange multipliers.

The proof of this proposition is found in Appendix A. This policy rule resembles a standard Taylor rule in its formulation. However, the nominal interest rate also reacts to the last period's output gap and current and past fiscal shocks in addition to the current inflation and output gap. It is possible to give an exact formula for the lag polynomial $\Xi(L)$, as presented in Appendix A - however, that is beyond the scope of this essay. This formula presents an analytic underpinning of the results of the following sections.

3 Parameter calibration

In estimating impulse response functions and the effect of different monetary policies, I used parameter estimates for the Eurozone found in Smets and Wouters (2003), Beck and Wieland (2010), Galí et al. (2001), and Casares (2001). I also took into account the estimates in Ireland (2004), which are based on US data. For determining the duration of long-term domestic government debt, I used data from OECD (2010) for the United States, backing out δ using the average yield-to-maturity in 2010. I used a baseline estimate for each parameter, while alternative estimates are considered in section 4. The values of the baseline parameter estimates are found in table 1.

The discount rate is, in most papers, assumed to be approximately 0.99 per period t. In Smets and Wouters (2003), β is equal to 0.99/quarter, equivalent to 0.9967/month and a steady-state yearly real interest rate $\overline{r}$ of 4%. Beck and Wieland (2010) do not consider the length of periods, but simply assume β to be equal to 0.99/period. Casares (2001), on the other hand, assumes and uses $\beta = 0.995$/quarter, which is equivalent to a discount rate of

Parameter	Value	Source	Interpretation
β	0.9967	Smets and Wouters (2003), Beck and Wieland (2010)	The discount factor of the consumer and the policymaker
$\overline{r}$	0.04	Smets and Wouters (2003), Beck and Wieland (2010)	The steady-state real interest rate
σ	1	Beck and Wieland (2010), Ireland (2004). Smets and Wouters (2003) use this as prior mean	The intertemporal elasticity of substitution
ϵ	6	Christiano et al. (2014)	The price elasticity of intermediate goods
ψ	2.503	Smets and Wouters (2003)	The Frisch elasticity of labour supply
η	0.428	Smets and Wouters (2003)	The elasticity of marginal cost with respect to firm-level production
θ	0.916	Casares (2001), Galí et al. (2001). Smets and Wouters (2003) use this as prior mean	The Calvo probability of no price change per period
δ	0.774	OECD (2010)	The theoretical coupon on the long-term governmental bonds

Table 1: Baseline parameter calibration

0.9983/month (or a 2% stead-state real interest rate), significantly less than the Smets and Wouters (2003) value. In this paper, I will use $\beta = 0.9967$/month (i.e. per period).

Beck and Wieland (2010) and Ireland (2004) assume an elasticity of intertemporal substitution equal to one, $\sigma = 1$, equivalent to log utilities. In contrast, Smets and Wouters (2003), using Bayesian maximum likelihood estimation, assuming a prior mean equal to 1, find $\sigma \approx 1.391$, while Casares (2001) finds $\sigma \approx 1.25$ using generalized method of moments estimation. Due to its simplicity and popularity, I will use the log utility specification as my baseline case. However, I will also analyse the implications of the Smets and Wouters and the Casares estimates.

As the price elasticity of intermediate goods, I use the standard value of $\epsilon = 6$, implying an average markup of 20%. However, using data from the post-war US economy, Andreasen and Dang (2019) find, using maximum likelihood estimation, an average markup on marginal

cost equal to 63% (or $\epsilon \approx 2.58$), significantly higher than my baseline estimate. Thus, I will also consider the effect of larger markups in my analysis, relying on this higher estimate.

The Frisch elasticity of labour supply was estimated to be $\psi \approx 2.503$ by Smets and Wouters (2003) - elasticities that are relatively high compared to the microeconomic estimates are standard in macroeconomic research, as Reichling and Whalen (2012) argued. Thus, I will use this as my baseline parameter value. As Keane and Rogerson (2012) argue, this difference between microeconomic and macroeconomic estimates might be due to several issues with the models commonly estimated in the microeconomic literature.

First, if human capital formation depends on hours worked, then the opportunity cost of time will depend on the effect of non-working on future income, not just on current wages. Then, the Frisch elasticity will be relatively low in the case of young workers but will increase with age, as gradual human capital accumulation reduces the importance of current work on future earnings. This would lead to a true elasticity closer to those estimated in the macroeconomic literature. Second, if the mapping from hours worked to earnings is nonlinear (in the case of Keane and Rogerson (2012), convexity is assumed), a linear regression specification will lead to underestimates of the Frisch elasticity compared to the true values.

Hence, I will use the estimated elasticity provided by Smets and Wouters (2003). However, Reichling and Whalen (2012) argue that macroeconomic estimates can be larger due to the effect of business cycles. If wages are sticky but unemployment fluctuates significantly, then a failure to control for cyclical effects will lead to the overestimation of the Frisch elasticity. Taking this into account, I will also consider the effects of a labour supply elasticity much lower than that of Smets and Wouters (2003) and closer to the standard microeconomic literature, $\psi = 0.4$, provided by Reichling and Whalen (2012).

For the elasticity of marginal cost with respect to firm-level production, I will use the widespread estimate of the capital share in income being $\alpha = 0.3$, giving us $\eta = 3/7 \approx 0.428$ from the formula $\eta = \alpha/(1-\alpha)$. Smets and Wouters (2003) also use this value for α in their framework.

The Calvo parameter of price stickiness is often assumed to be $\theta = 0.75$/quarter, or

≈ 0.916/month - Casares (2001) uses this value - and Galí et al. (2001) also estimated, using GMM, a value between 0.699 and 0.771 for the Eurozone. I will also use this as my baseline estimate. However, Smets and Wouters (2003), using this value as the prior mean, estimated, using Bayesian methods, $\theta \approx 0.9683$/month. Thus, I will use a higher value, 0.999, as an alternative parameter value in my analysis, checking the implications of significantly stickier prices. In addition, I will analyse the effects of highly flexible prices, given by $\theta = 0.1$.

I estimated the theoretical infinite-period bonds' coupon value using US data from OECD (2010), indicating a Macaulay duration of US domestic government debt equal to $D_t = 4$ in 2010. From Federal Reserve Bank of St. Louis (2022) data, the approximate yearly yield on US 10-year treasury bonds in 2010 was $1 + j_t \approx 1.0321$. Using the formula $D_t = (1+j_t)/(1+j_t-\delta)$, we can get the estimated value of the theoretical coupon of $\delta \approx 0.774$. I will use this as my baseline value, even though this is corresponding to a yearly coupon - using it with monthly data implies that about 23% of all debt is refinanced each month in steady state - as it can give a better picture of the behaviour of less developed European economies, relying on shorter maturity debt. In a crisis, when fiscal shocks become likelier, short debt durations can be especially relevant. However, for completeness, I will look at alternative coupon values and durations in my analysis, partly based on Doepke and Schneider (2006).

4 Discussion of results

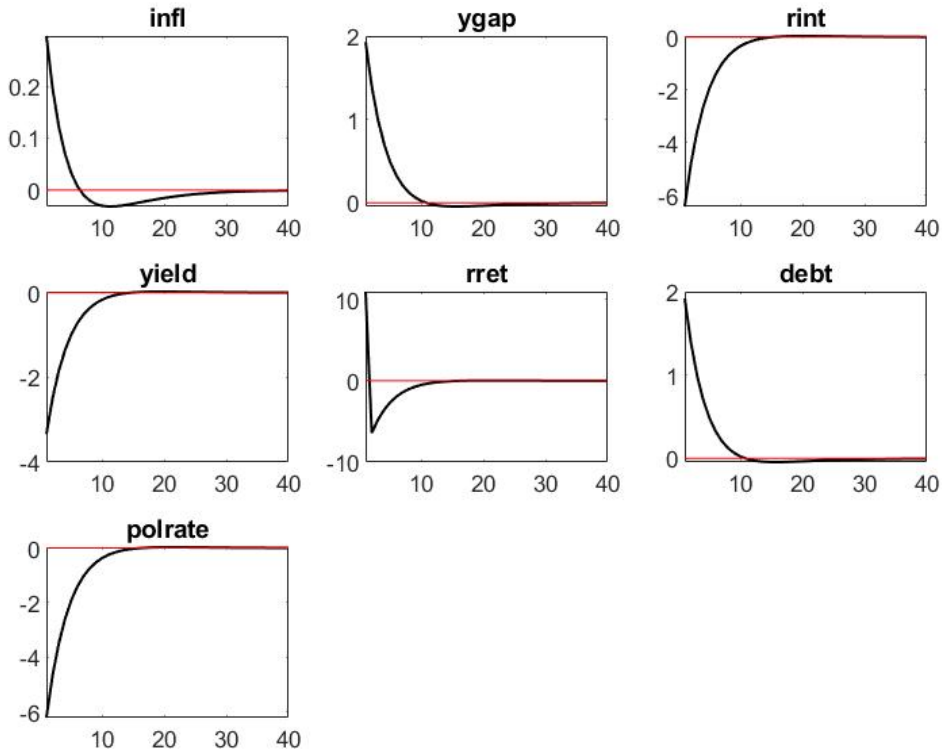
4.1 Baseline case

First, I will look at the baseline model with different parameter values. Throughout this section, I will analyse the variance and path of the macroeconomic variables in the model responding to the fiscal shock, under optimal Ramsey policy. The shock has a value equal to the standard deviation of the fiscal policy random variable, equal to 0.01. As the model is entirely linear, the absolute value of the shock does not matter in the sense that impulse response functions are just scaled up if the shock increases, *ceteris paribus*. The simulation

will span 100 time periods, and I will also be looking at the value of the welfare loss, which is the sum of the values of the loss function throughout the simulation, discounted with the policymaker's discount factor.

The path of the variables in the baseline calibration with optimal policy is seen in figure 1, shown up to only 40 periods for clarity. Increasing government expenses are covered by raising real long-term debt. However, as future taxes cannot compensate for unexpected spending shocks, the real return on bonds must be reduced to negative levels to force real debt back to steady-state. A reduction in the nominal interest rate and the yield-to-maturity of government bonds is the principal instrument of the policymaker in order to achieve this - a loose monetary policy leads to inflation, which further reduces the real interest rate and the real return.

Figure 1: The baseline case



(infl denotes inflation, ygap denotes output gap, rint denotes the short-term real interest rate, yield denotes the yield-to-maturity, rret denotes the ex-post real return on long-term bonds, debt denotes real government debt, and polrate is the policy rate, which is the short-term nominal interest rate.)

Real returns jump significantly in the period of the shock - this is due to the fact that a reduction in yields boosts bond prices, allowing bondholders to sell their assets at a price significantly higher than expected initially. The government's debt repayment costs increase in the first period; hence the value of the real debt rises by more than the value of the fiscal shock. Financial repression starts putting downward pressure on debt only from the second time period onward when returns fall below steady-state.

It is clear that the rise in inflation is significantly smaller than the rise in the output gap, implying a focus on price stability by the policymaker - in this sense, our results are similar to those of Schmitt-Grohe and Uribe (2002) and Siu (2004). However, inflation is traded off for changes in output gap, not higher income taxes in our case.

This "inflation stabilization bias" is fundamentally explained by the fact that nominal and firm-level real rigidities are high under the baseline calibration. Coupled with a relatively elastic demand schedule, this implies that inflation can lead to severe distortions in the distribution of individual goods in the consumption basket, relative to the efficient distribution, under no inflation.[4] The sluggishness of price changes affects welfare through an insufficient accommodation of the significantly price-sensitive household preferences.

Aggregate-level real rigidities are also relatively high, which means that marginal costs have a relatively muted response to changes in the output gap. In addition, a high Frisch elasticity of labour supply reflects a relatively less significant welfare loss resulting from extra hours worked supporting a positive output gap - hence, the relative importance of an output gap deviation relative to inflation is reduced.

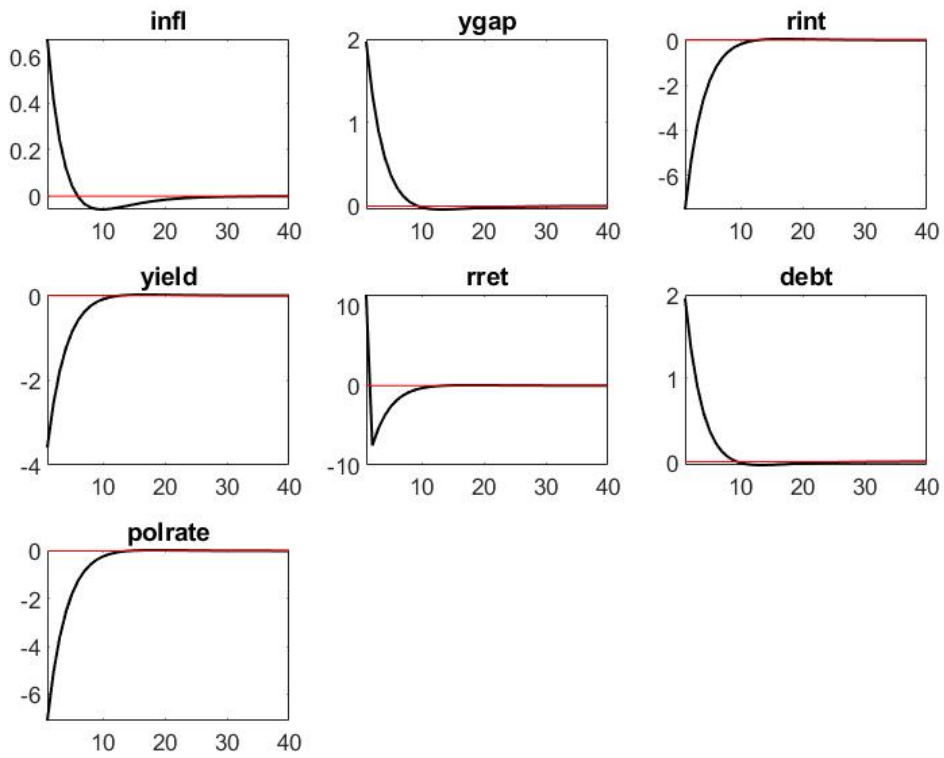
These factors are reflected in the coefficients on the square of inflation and output gap in the welfare loss function: that on inflation is approximately 1400 times larger than that on output. The difference in the path of these two variables is not as drastic due to the form of the loss function, and that the correlation between the two is high due to the structure of the model. Hence, trade-offs are sub-optimal above some moderate level.

The present value of the welfare loss equals approximately 0.09% of the value of steady-state consumption - the absolute value is not important, as larger shocks would result in

[4]The efficient distribution with the Dixit-Stiglitz aggregator is characterized by the equal consumption of all individual goods, implying equal prices.

proportionally higher losses. However, it provides a good reference for the comparison of welfare losses under different calibrations.

Figure 2: Lower elasticity of labour supply



Using a Frisch elasticity of 0.4, based on Reichling and Whalen (2012), the representative household's welfare loss increases under the optimal policy. The simple direct reason is that the disutility of hours worked is more considerable; hence a positive deviation from the steady-state of output is necessarily more costly in consumption terms, *ceteris paribus*.

The indirect mechanism is that aggregate-level real rigidities are reduced so that changes in the output gap are passed over at a higher rate into marginal costs. Hence, the correlation between inflation and output is more significant, reducing trade-off opportunities between the two factors. This indirect effect is reflected by the higher relative importance of output deviations to inflation deviations in the welfare loss function - the price level rises more relative to the output gap than in the baseline case. Overall, both variables reach a higher initial level than in the baseline case.

Intuitively, as output increases, workers have to be compensated more generously to in-

duce them to supply more hours - leading to both higher marginal costs (inducing inflation) and utility loss from extra work. Overall, these effects amount to a welfare loss, with present value being almost three times as high as in the baseline case.

Relative to the baseline case, a 40% higher intertemporal elasticity of substitution - based partly on the estimates of Smets and Wouters (2003) and Casares (2001) - also leads to a higher welfare loss, almost twice as big. The inflation and output variance is higher, both reaching higher maxima, while debt and interest rates exhibit lower variability.

A higher IES leads to higher aggregate-level real rigidities and reduced importance of the output gap in the loss function. However, consumption and output become more sensitive to interest rate changes. Thus, the policymaker aims to minimise interest rate deviations but is constrained by the limiting condition of the real debt.

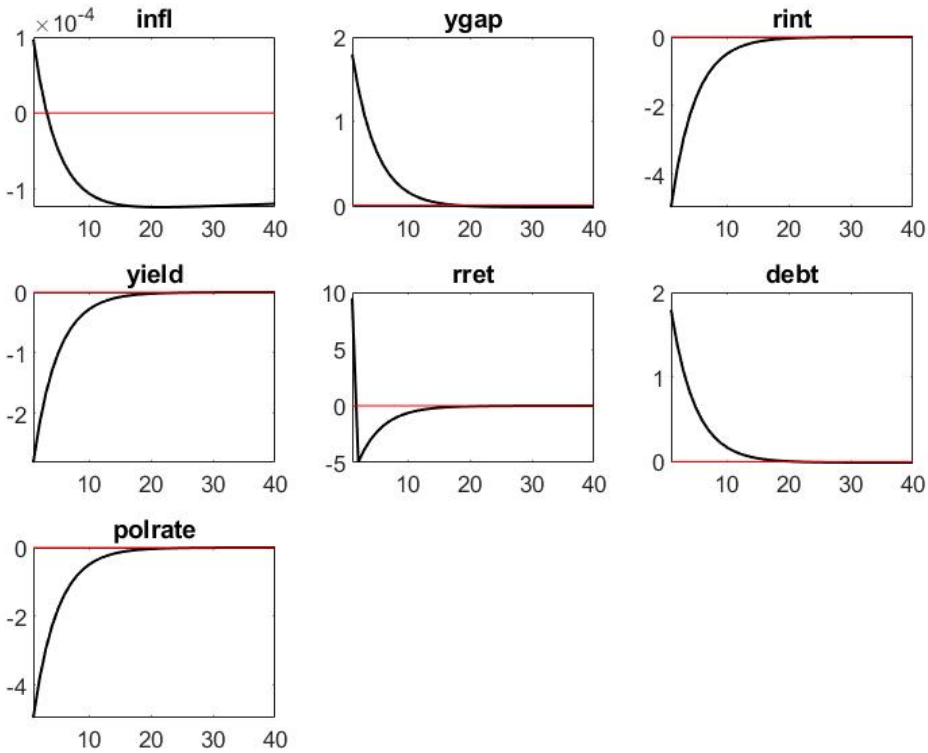
In contrast, a surprising result of this model is that a less elastic demand for intermediate goods of 2.58 (based on Andreasen and Dang (2019)), leading to markups increasing by more than 40 percentage points, will reduce overall welfare loss from 0.09% to 0.079% of the value of steady-state consumption. Financial repression through inflation is less costly due to the lower sensitivity of consumption demand. That is, households do not change the relative proportion of given individual goods in their basket by as much as in the baseline scenario responding to a rise in price dispersion, and the distribution of the goods in the basket remains closer to the efficient distribution of complete equality.

Even though firm-level real rigidities fall, allowing average marginal costs to pass through more easily to firm prices, the reduced demand elasticity is powerful enough to counteract the detrimental influence of inflationary pressure. This is reflected in the fact that, as the output gap becomes relatively more important for the welfare of the representative household, the policymaker allows inflation to rise by more, curbing output deviations relative to the baseline scenario.

Similarly, changes in price stickiness can have significant welfare-improving effects. If prices become extremely sticky ($\theta = 0.999$), financial repression through inflation would be extremely costly in consumption terms, as price distortions would have a high persistence. However, as the results from Section 2 confirm, inflation is very unresponsive to the policy

rate and output gap due to greater nominal rigidities. This allows the policymaker to focus on the trade-off between real debt reduction and output gap deviations and reduce welfare loss to approximately 0.078% of steady-state consumption. Intuitively, firms have a hard time changing prices in the first place, so a temporary shock is not particularly important.

Figure 3: Calvo parameter close to unity - very sticky prices

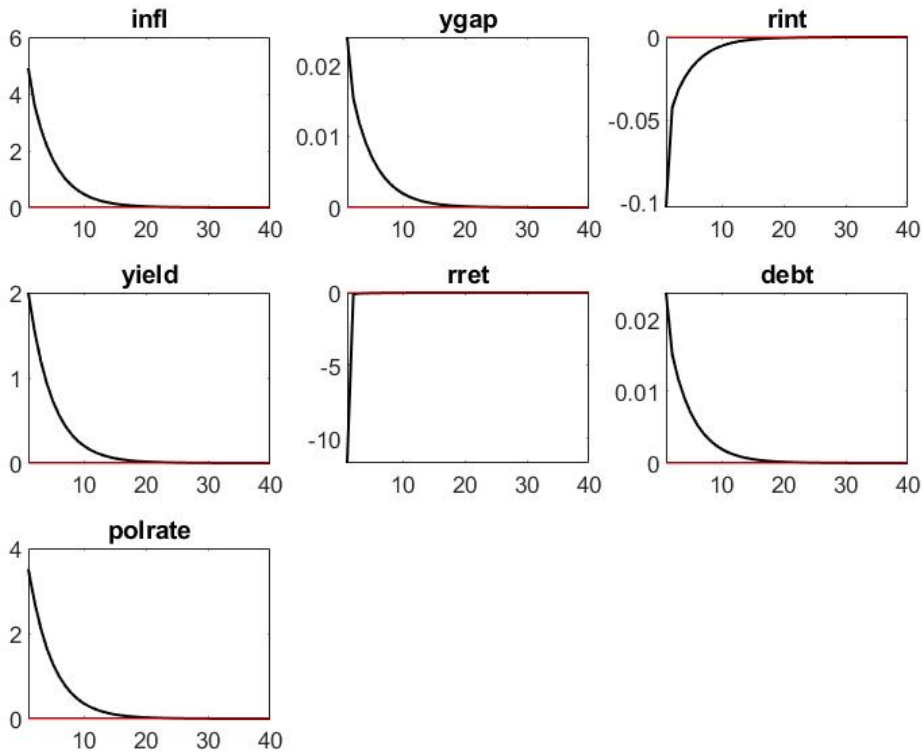


Approaching complete price flexibility - with $\theta = 0.1$ - welfare loss becomes negligibly small. Inflation becomes a much more useful tool for financial repression, as firms can adjust their prices much more easily, smoothing distortions and leading to a muted response of output to increased spending in any given period - as seen in figure 4.

Nominal rigidities become so insignificant that a higher nominal interest rate will now have a positive current effect on both inflation and the output gap. Thus, the policymaker will have to increase rates to reduce real returns through inflation, which counteracts the effect of the shock on real debt almost instantly, keeping it approximately one hundred times lower than in the previous scenarios.

The duration and coupon value of the government bonds also affect the effectiveness of

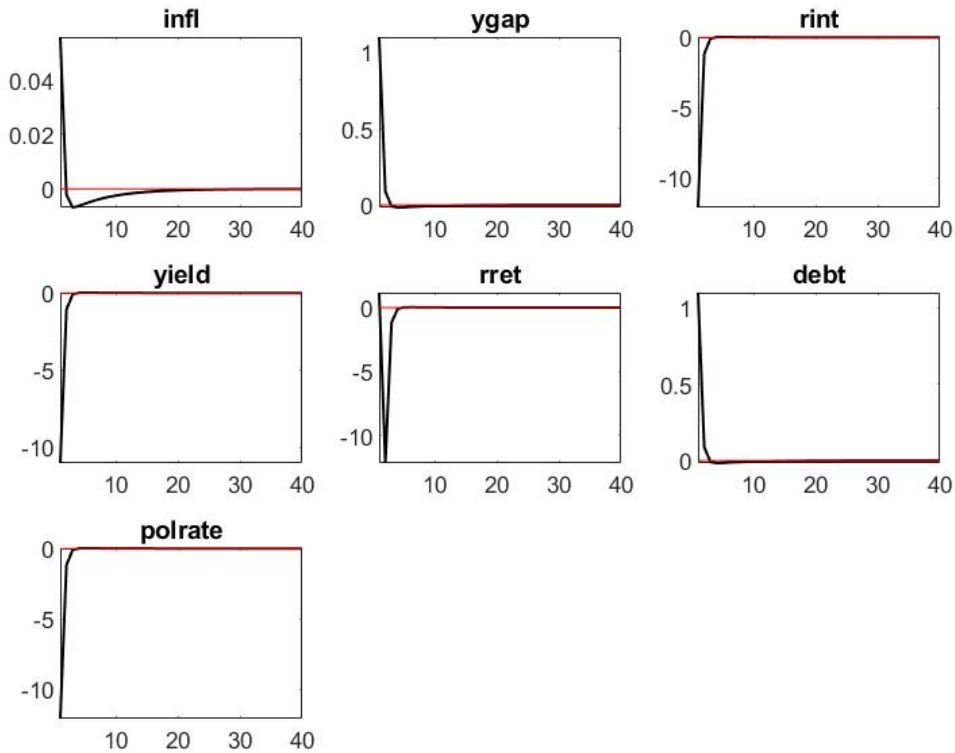
Figure 4: Calvo parameter close to zero - very flexible prices



financial repression: both very short and very long durations aid efforts to reduce real debt. A low coupon value - implying a short duration of debt - allows the government to cut the value of the real debt almost instantly, as seen in figure 5, reducing welfare loss to a ninth of its baseline value by precluding significant long-term effects. Inflation does not play a significant role in this scenario, as the nominal interest rate cut is reversed almost entirely in the next period, preventing future expectations from exerting upward inflationary pressure.

In contrast, a high coupon value leads to persistent but very limited real effects, as seen in figure 6, which does underscore the relevance of the findings of Lustig et al. (2008) in our model, although not for the same reason as in their paper. The real interest rate and the real return still achieve most of the debt reduction in the first periods. However, now this is achieved by a long-run nominal rate rise, as higher nominal interest rates positively affect inflation. The policymaker reduces the present value of the loss by exploiting the high duration of its debts and postponing the consequences on the real economy into the far future, which is discounted more heavily by the representative household. Similarly to the

Figure 5: Low debt duration



case with high price flexibility, inflation is a more prominent factor in the optimal financial repression strategy than output gap volatility.

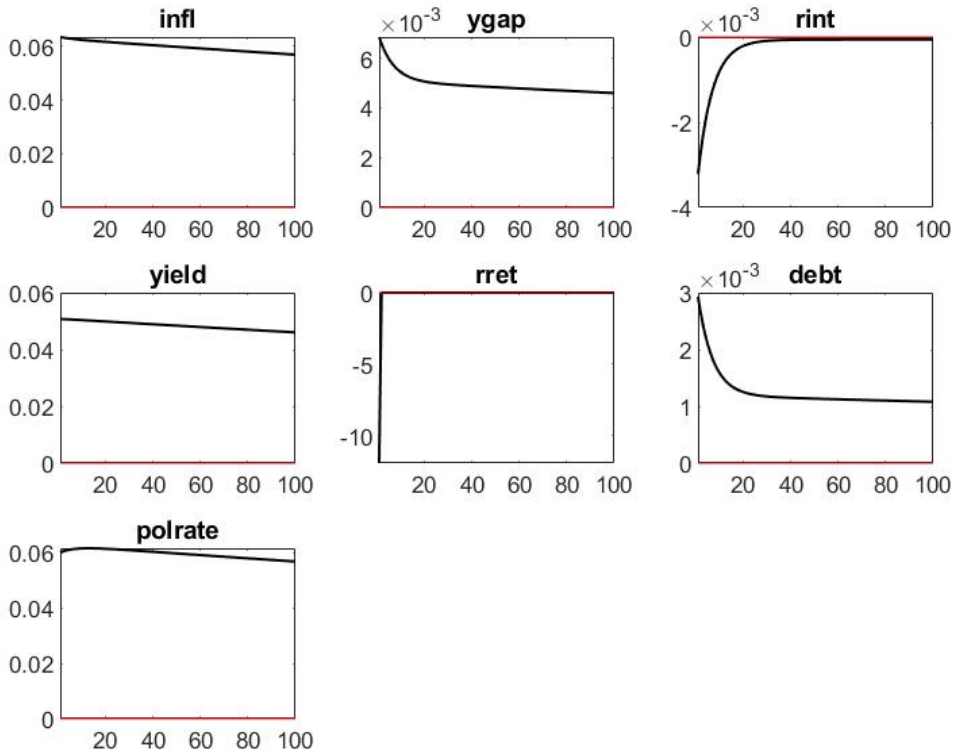
Overall, the emphasis of Schmitt-Grohe and Uribe (2002) and Siu (2004) on the importance of price stability holds in our model, too - it is much more desirable to use deviations in output gap in order to reduce real debt to a stable level than to inflate it away. If prices are extremely flexible (or the duration of the government debt is very high), this conclusion does not hold anymore - the primary reason why price stability has salutary effects is the market inefficiency introduced by price stickiness.

4.2 Persistent fiscal shocks

Now, assume that the fiscal shock follows a simple AR(1) process:

$$\phi_t = \rho\phi_{t-1} + \upsilon_t$$

Figure 6: High debt duration

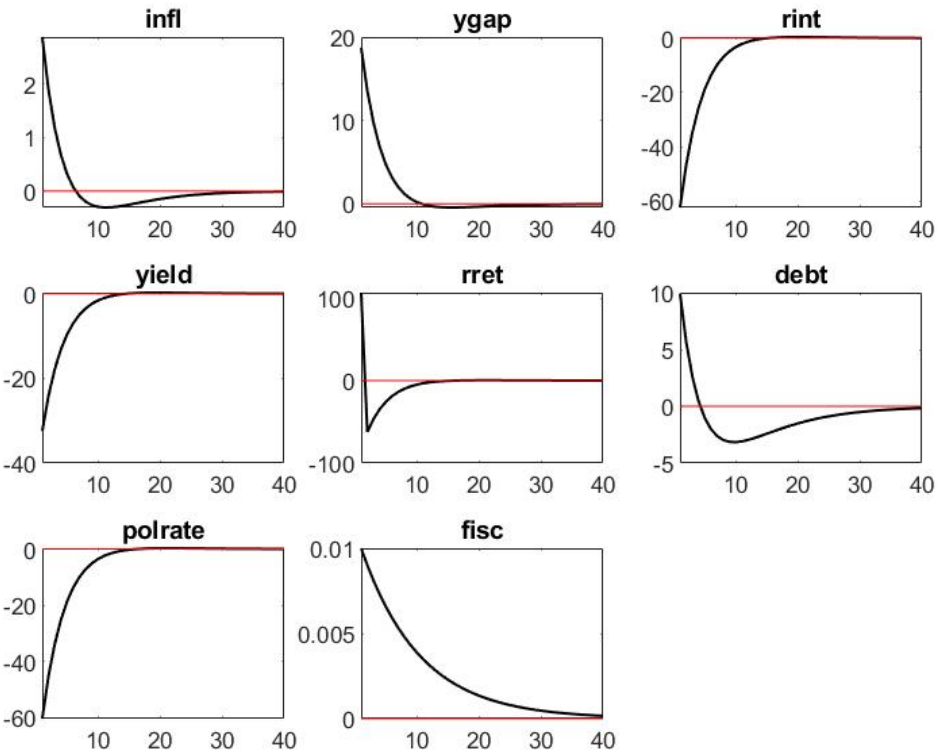


Where υ is an i.i.d. white noise process and ρ is the persistence of the shock. A persistent fiscal shock is analogous to a government expenditure path in an extended political or economic crisis, such as a war or a protracted pandemic: a considerable mobilization of funds and investments is followed by a gradual reduction of expenditures.

Note that, for example, this is a good approximation for the period of the financial crisis in the United States after 2008, but spending slopes upwards in the first years of wars usually, after which it is phased out, as can be seen from the data of The Office of Management and Budget of the United States (2022). In addition, taxes are much likelier to be used to offset expenditures in wars, as the experience of the United States in the Second World War shows, with large increases in the taxes of higher income percentiles. Nevertheless, financial repression was an essential part of the post-war policy toolkit aiming to reduce war debt under the Bretton Woods system, as Reinhart and Sbrancia (2011) argue.

A persistence of 0.9 - which is not particularly high for monthly data, implying an overall fiscal shock value of 0.1 for a first period shock of 0.01 - leads to an almost one hundred times

Figure 7: Persistent fiscal shock



larger welfare loss than in the baseline case of 8.51%. This is still lower than a one-period fiscal shock of 0.1 in the baseline case, where the loss is 9.21% of steady-state consumption - the loss caused by a smaller but persistent shock is lower because the government has more time to accommodate it in the future, which is also discounted.

As seen in figure 7, the path of the real variables is similar to the baseline case, with the exception that the deviations from steady-state are approximately ten times larger. Real debt becomes much more volatile, and the policymaker is forced to reduce the nominal rate by an unreasonable (due to the possible existence of a lower bound) 60 percentage points below steady-state.

4.3 Simulation with an effective lower bound

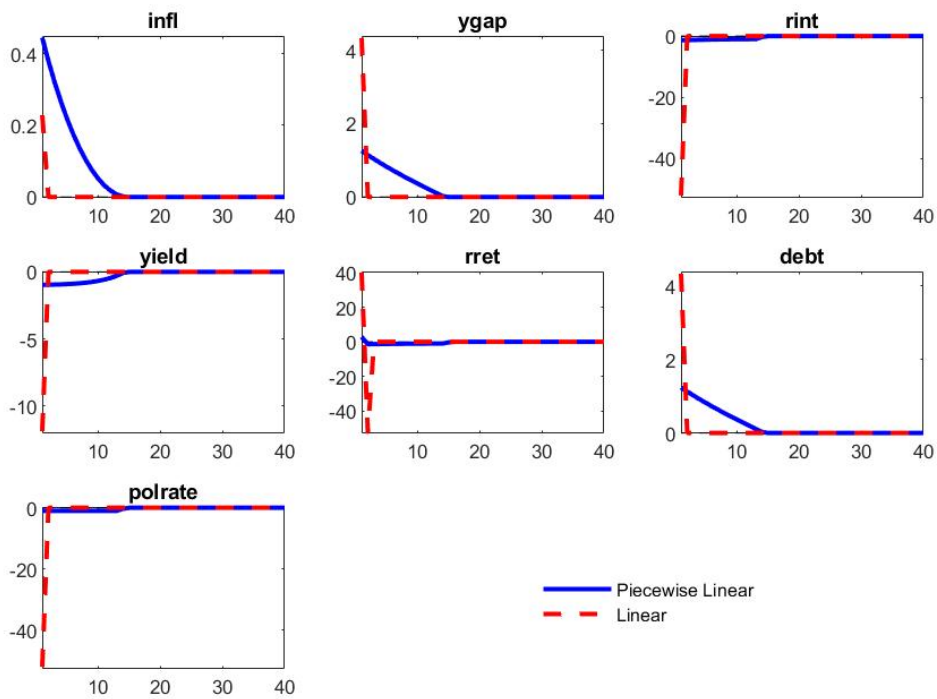
Here, we look at the consequences on policymaking of an effective lower bound of one percentage point below steady-state. As the effective lower bound constrains the government,

we would expect that it leads to a higher welfare loss. This is indeed the case under optimal policy - the Ramsey policymaker's actions are further constrained. However, this conclusion is not necessarily true if we assume that the policymaker follows a simpler, more realistic rule. Let us assume that the monetary authority follows:

$$i_t = -d_t$$

That is, it aims to reduce the monthly nominal cost of short term borrowing one-to-one whenever the level of debt increases. In the short run, this function can be a good approximation of the behaviour of a government aiming to suppress the effects of an expenditure shock - in the long run, other variables also have to be taken into account. Then, we assume here that the effective lower bound binds at $i_t = -1\%$

Figure 8: Non-optimal policy under the effective lower bound



(The blue, piecewise linear IRFs present the paths if an ELB exists at -1%, while the red, linear path presents the path without an ELB.)

The path of the variables are seen in figure 8. The response of the variables is muted when there exists a lower bound, as the policymaker cannot reduce the nominal rate by as

much as the rule would dictate. Inflation rises due to the forward guidance effect, as the policy rate is below steady-state for longer than in the unconstrained case.

The policy rule is not optimal, and the policy response with no lower bound results in a welfare loss greater than under an optimal rule. The deviations around the steady-state, especially that of the output gap, are too drastic, as the government aims to reduce real debt to its steady-state as quickly as possible, virtually achieving this by the second period (in one month).

The effective lower bound prevents this by moderating the possible size of policy rate deviations, forcing the monetary authority to follow a much smoother path: this reduces welfare loss relative to the unconstrained case by 32%. This suggests that the effective lower bound can have significantly beneficial effects by preventing excessively extreme financial repression if the government is not sensitive enough to household welfare.

5 Conclusion

This essay proposed a framework and several results concerning optimal policy and financial repression in a simple New Keynesian model, augmented with a government sector and a long-term bond market. Theoretical results concerning the policymaker's behaviour underscore how it can trade off output and inflation, the two components of the household welfare loss function. I also discussed the concrete form of the optimal policy response function under our model specification, similar to a usual targeting rule for the nominal interest rate responding to output gap and inflation.

I provided a detailed analysis of the model variables' behaviour under the optimal policy regime reacting to a fiscal shock - notably, the essence of the conclusions of Schmitt-Grohe and Uribe (2002) and Siu (2004) concerning the importance of price stability in fiscal policy held under most model specifications. Price flexibility limits this, whereby inflation becomes a valuable debt reduction tool again. The government debt duration also plays an important part - as I have argued, both very long and very short durations can have salutary welfare effects.

In addition, I discussed the implications of shock persistence and an effective lower bound. A large, instantaneous shock is much more harmful from a household welfare point of view than a smaller, persistent shock with the same total value through time periods, simply because the policymaker has more time to accommodate the shock in the future.

In contrast, the presence of an effective lower bound on the short-term nominal interest rate must reduce the scope of the optimal policymaker's actions, as it introduces an additional constraint, aggravating welfare loss. However, in the presence of myopia and non-optimal policy, a lower bound can prevent the government from executing extreme policy plans, possibly reducing welfare loss by as much as 32%.

Further research involving more complicated models - including, importantly, income taxation - has the potential to check the conclusions of this essay and understanding the scope of its relevance. It is essential that we understand the construction and the implications of optimal policy and optimal debt reduction, especially in the current climate of high inflation and large sovereign debts.

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Appendix A

The solution to the optimal Ramsey policy problem

The Ramsey policymaker minimizes the following loss function in period t_0:

$$L_{t_0} = \frac{1}{2}\sum_{t=t_0}^{\infty} \beta^{t-t_0}\mathbb{E}_{t_0}\left[\frac{\epsilon}{\nu\xi}\pi_t^2 + \gamma y_t^2\right]$$

Subject to the following constraints:

$$\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa y_t$$

$$y_t = \mathbb{E}_t y_{t+1} - \sigma r_t$$

$$r_t = \mathbb{E}_t R_{t+1}$$

$$j_t = \beta\delta\mathbb{E}_t j_{t+1} + (1-\beta\delta)i_t$$

$$R_t = j_{t-1} - \pi_t - \frac{\beta\delta}{(1-\beta\delta)}(j_t - j_{t-1})$$

$$\beta d_t = d_{t-1} + R_t + \beta\phi_t$$

And the transversality conditions. In order to solve this, we can define the following Lagrangian function:

$$
\begin{aligned}
\Lambda_{t_0} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} \mathbb{E}_{t_0} \Bigg[& \frac{1}{2}\left(\frac{\epsilon}{\nu\xi}\pi_t^2 + \gamma y_t^2\right) \\
& - \lambda_t \left(\pi_t - \beta\pi_{t+1} - \kappa y_t\right) \\
& - \mu_t \left(y_t - y_{t+1} + \sigma r_t\right) \\
& - \Omega_t \left(r_t - R_{t+1}\right) \\
& - \tau_t \left(j_t - \beta\delta j_{t+1} - (1-\beta\delta) i_t\right) \\
& - \omega_t \left(R_t - j_{t-1} + \pi_t + \frac{\beta\delta}{(1-\beta\delta)}(j_t - j_{t-1})\right) \\
& - \zeta_t \left(\beta d_t - d_{t-1} - R_t - \beta\phi_t\right) \Bigg]
\end{aligned}
$$

Solving for the maximizing values of the endogenous variables, we find the following first order conditions:

$$\frac{\epsilon}{\nu\xi}\pi_t - \lambda_t + \lambda_{t-1} - \omega_t = 0 \tag{1}$$

$$\gamma y_t + \kappa\lambda_t - \mu_t + \frac{1}{\beta}\mu_{t-1} = 0 \tag{2}$$

$$\sigma\mu_t + \Omega_t = 0 \tag{3}$$

$$\frac{1}{\beta}\Omega_{t-1} - \omega_t + \zeta_t = 0 \tag{4}$$

$$-\tau_t + \delta\tau_{t-1} + \frac{\beta}{(1-\beta\delta)}\mathbb{E}_t\omega_{t+1} - \frac{\beta\delta}{(1-\beta\delta)}\omega_t = 0 \tag{5}$$

$$(1-\beta\delta)\tau_t = 0 \tag{6}$$

$$\mathbb{E}_t\zeta_{t+1} - \zeta_t = 0 \tag{7}$$

I assume that the economy starts from steady state, meaning that $\pi_t = y_t = 0, \forall t < t_0$. This means that the value of the Lagrange multipliers is equal to zero, $\forall t < t_0$, and thus we can infer from equation (6) that $\tau_t = 0, \forall t$. This mirrors the fact that the policy rate is unconstrained by equations other than the bond yield equation, allowing the policymaker to satisfy this constraint for any value of expected and current yield-to-maturity. Hence, (5)

can be rearranged as:

$$\mathbb{E}_t \omega_{t+1} = \delta \omega_t, \ \forall t \tag{8}$$

Substituting (7) into (4) and rearranging gives:

$$\mathbb{E}_t \omega_{t+1} - \frac{1}{\beta}\Omega_t = \omega_t - \frac{1}{\beta}\Omega_{t-1}$$

Substituting in equation (8) yields the following second order difference equation:

$$(1+\delta)\Omega_t = \mathbb{E}_t \Omega_{t+1} + \delta \Omega_{t-1}$$

A solution of this stochastic equation is:

$$\Omega_t = (1+\delta)\Omega_{t-1} - \delta \Omega_{t-2} + \Theta(L)\phi_t$$

Where $\Theta(L)$ is a lag polynomial, L being the lag operator. Using this, we can get from equation (3) that μ_t follows the process:

$$\mu_t = (1+\delta)\mu_{t-1} - \delta \mu_{t-2} - \frac{1}{\sigma}\Theta(L)\phi_t$$

By combining this process with equation (2), it can be shown that the following identity holds:

$$(1+\delta)\beta\gamma y_t - \beta\gamma \mathbb{E}_t y_{t+1} - \delta\beta\gamma y_{t-1} = -(1+\delta)\beta\kappa\lambda_t + \beta\kappa\mathbb{E}_t\lambda_{t+1} + \delta\beta\kappa\lambda_{t-1} - \frac{1}{\sigma}\Theta(L)\phi_t \tag{9}$$

Substituting equation (8) into (1), we will get:

$$\frac{\epsilon}{\nu\xi}\left(\pi_t - \frac{1}{\delta}\mathbb{E}_t\pi_{t+1}\right) = \frac{1+\delta}{\delta}\lambda_t - \lambda_{t-1} - \frac{1}{\delta}\mathbb{E}_t\lambda_{t+1}$$

Substituting in (9) into this equation gives us the following identity:

$$\frac{\epsilon}{\nu\xi}\left(\delta\pi_t - \mathbb{E}_t\pi_{t+1}\right) = \frac{\gamma}{\kappa}\mathbb{E}_t y_{t+1} - \frac{(1+\delta)\gamma}{\kappa}y_t + \frac{\gamma\delta}{\kappa}y_{t-1} - \frac{1}{\sigma\beta\kappa}\Theta(L)\phi_t$$

Substituting in the Phillips curve and the IS curve for the expectation of inflation and the output gap, we get:

$$\epsilon(\beta\delta - 1)\pi_t = \frac{\nu\xi\beta\gamma\sigma}{\kappa}r_t - \frac{\epsilon\kappa^2 + \nu\xi\beta\delta\gamma}{\kappa}y_t + \frac{\nu\xi\beta\gamma\sigma}{\kappa}y_{t-1} - \frac{\nu\xi}{\sigma\kappa}\Theta(L)\phi_t$$

By using the no-arbitrage condition on the real interest rate and the expected real returns, we get:

$$\epsilon(\beta\delta - 1)\pi_t = \frac{\nu\xi\beta\gamma\sigma}{\kappa}\mathbb{E}_t R_{t+1} - \frac{\epsilon\kappa^2 + \nu\xi\beta\delta\gamma}{\kappa}y_t + \frac{\nu\xi\beta\gamma\sigma}{\kappa}y_{t-1} - \frac{\nu\xi}{\sigma\kappa}\Theta(L)\phi_t \tag{10}$$

All that remains is to find an expression for $\mathbb{E}_t R_{t+1}$ in current and lagged variables in order to find the solution. By using the equations for the ex-post return on bonds and for the bond yield, we get that:

$$\mathbb{E}_t R_{t+1} = i_t - \frac{1}{\beta}\pi_t + \frac{\kappa}{\beta}y_t$$

By substituting this in equation (10), we will get:

$$\Big[\frac{\kappa\epsilon(\beta\delta - 1)}{\nu\xi\gamma\sigma} + 1\Big]\pi_t = \beta i_t + \Big(\kappa - \frac{\epsilon\kappa^2 + \nu\xi\beta\delta\gamma}{\nu\xi\gamma\sigma}\Big)y_t + \beta y_{t-1} - \frac{1}{\gamma\sigma^2}\Theta(L)\phi_t$$

Which, denoting $\frac{1}{\gamma\sigma^2}\Theta(L) = \Xi(L)$, gives us the policy rule in standard form:

$$i_t = \frac{1}{\beta}\Big[\frac{\kappa\epsilon(\beta\delta - 1)}{\nu\xi\gamma\sigma} + 1\Big]\pi_t - \frac{1}{\beta}\Big(\kappa - \frac{\epsilon\kappa^2 + \nu\xi\beta\delta\gamma}{\nu\xi\gamma\sigma}\Big)y_t - y_{t-1} + \Xi(L)\phi_t$$

□

Appendix B

The inflation - output gap trade-off

We assume perfect foresight and full commitment throughout. The monetary authority can affect inflation and the output gap by committing to current and future changes in the nominal short-term interest rate, taking into account the Phillips curve and the Investment-Savings curve (where the Fisher equation can be substituted in for the real interest rate):

$$\pi_t = \beta\pi_{t+1} + \kappa y_t$$

$$y_t = y_{t+1} - \sigma(i_t - \pi_{t+1})$$

Iterating these forward gives us:

$$\pi_t = \kappa \sum_{l=0}^{\infty} \beta^l y_{t+l} + \lim_{L\to\infty} \beta^L \pi_{t+L}$$

$$y_t = -\sigma \sum_{l=0}^{\infty} (i_{t+l} - \pi_{t+l+1}) + \lim_{L\to\infty} y_{t+L}$$

Imposing that the output gap and inflation return to steady-state, we get:

$$\pi_t = \kappa \sum_{l=0}^{\infty} \beta^l y_{t+l} \tag{11}$$

$$y_t = -\sigma \sum_{l=0}^{\infty} (i_{t+l} - \pi_{t+l+1}) \tag{12}$$

I assume that the future nominal interest rates are independent (there is no monetary policy function). Substituting (11) in (12) and differentiating with respect to i_{t+m} we will get:

$$\frac{dy_t}{di_{t+m}} = -\sigma + \sigma\kappa \sum_{l=0}^{\infty}\sum_{j=0}^{\infty} \beta^j \frac{dy_{t+l+j+1}}{di_{t+m}}$$

Based on the previous assumptions and as both inflation and the output gap are forward-looking variables, $\frac{dy_{t+i}}{di_t} = 0$ and $\frac{d\pi_{t+i}}{di_t} = 0, \forall i > 0$, so we can write:

$$\frac{dy_t}{di_{t+m}} = -\sigma + \sigma\kappa \sum_{l=0}^{m-1}\sum_{j=0}^{m-l-1} \beta^j \frac{dy_{t+l+j+1}}{di_{t+m}}$$

The summation can be rewritten in the following way:

$$\frac{dy_t}{di_{t+m}} = -\sigma + \frac{\sigma\kappa}{1-\beta} \sum_{l=1}^{m}(1-\beta^l)\frac{dy_{t+l}}{di_{t+m}}$$

Defining $g\colon \mathbb{N} \to \mathbb{R}$, $g(m) = \frac{dy_t}{di_{t+m}}$, where $g(0) = -\sigma$, we can write:

$$g(m) = -\sigma + \frac{\sigma\kappa}{1-\beta} \sum_{l=1}^{m}(1-\beta^l)g(m-l)$$

Now, define:

$$G(z) \equiv \sum_{s=0}^{\infty} g(s)z^s$$

We can find $G(z)$ by using the definition of $g(m)$:

$$\begin{aligned}
G(z) &= \frac{-\sigma}{1-z} + \frac{\sigma\kappa}{1-\beta} \sum_{s=1}^{\infty}\sum_{l=1}^{s}(1-\beta^l)g(s-l)z^s \\
&= \frac{-\sigma}{1-z} + \frac{\sigma\kappa}{1-\beta} \sum_{k=0}^{\infty} g(k)z^k \sum_{l=1}^{\infty}(1-\beta^l)z^l \\
&= \frac{-\sigma}{1-z} + \frac{\sigma\kappa}{1-\beta} G(z) \sum_{l=1}^{\infty}(1-\beta^l)z^l
\end{aligned}$$

Which reduces to:

$$G(z) = \frac{\sigma\beta z - \sigma}{1-(1+\beta+\sigma\kappa)z + \beta z^2} = \frac{c_1}{z-r_1} + \frac{c_2}{z-r_2}$$

Where:

$$r_1 = \frac{\frac{(1+\beta+\sigma\kappa)}{\beta} + \sqrt{\frac{(1+\beta+\sigma\kappa)^2}{\beta^2} - \frac{4}{\beta}}}{2}$$
$$r_2 = \frac{\frac{(1+\beta+\sigma\kappa)}{\beta} - \sqrt{\frac{(1+\beta+\sigma\kappa)^2}{\beta^2} - \frac{4}{\beta}}}{2}$$
$$c_1 = \frac{\frac{\sigma}{\beta} - \sigma r_1}{r_2 - r_1}$$
$$c_2 = \frac{\sigma r_2 - \frac{\sigma}{\beta}}{r_2 - r_1}$$

It is clear from the definition of $G(z)$ that [5]:

$$\frac{G^{(m)}(0)}{m!} = g(m) = \frac{dy_t}{di_{t+m}} \tag{13}$$

Where $G^{(m)}(z)$ denotes the m-th derivative of the function G. It can be easily shown that the general formula for g(m) is, then:

$$g(m) = (-1)^{m+1}\left(\frac{c_1}{r_1^{m+1}} + \frac{c_2}{r_2^{m+1}}\right)$$

Differentiating equation (11) with respect to i_{t+m} and substituting in (13) yields:

$$\frac{d\pi_t}{di_{t+m}} = \kappa \sum_{l=0}^{m} \beta^l g(m-l)$$

And it is clear that the change in inflation relative to the change in output gap will be a function of the distance from the change in the policy rate:

$$\left.\frac{d\pi_t}{dy_t}\right|_{di_{t+m}} = \kappa \sum_{l=0}^{m} \beta^l \frac{g(m-l)}{g(m)}$$

□

[5] I thank Professor Robert Israel (University of British Columbia) for pointing out the possibility of this transformation.